

HUJI Applicative Research Report — Exact & Social Sciences — October 2026

2 paper(s) · Exact & Social Sciences · Monthly · October 2026 · Generated October 01, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

This month's featured research highlights two significant advancements in materials science for clean energy applications. One project provides fundamental insights into sodium-ion battery mechanisms, paving the way for more cost-effective and sustainable grid-scale energy storage solutions. The other unravels the complexities of catalyst degradation, offering a novel pathway to develop more durable and self-healing catalysts crucial for high-efficiency hydrogen production and fuel cell technologies. These innovations represent vital steps towards a more sustainable and efficient energy future.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- Daphna Shimon — Materials, Clean Energy
- Elad Gross — Materials, Clean Energy

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Elucidating Sodium-Ion Storage Mechanisms in Bowl-Shaped Hierarchical Porous Carbon Anodes: A Combined Solid-State NMR and EPR Approach

23/50

"Unlocking cheaper, sustainable grid-scale energy storage with advanced sodium-ion battery materials."

Main researcher: Daphna Shimon · daphna.shimon@mail.huji.ac.il

Institute of Chemistry , The Hebrew University of Jerusalem , ,

Published · 2026 Sep 26

ABOUT THIS RESEARCH

This research uses advanced spectroscopic techniques to understand how sodium ions move and store energy within specialized porous carbon structures. By mapping these microscopic mechanisms, researchers can better design high-performance electrodes for next-generation battery technologies.

COMMERCIAL ANGLE

The ability to engineer hierarchical porous carbons provides a pathway to developing cheaper, more sustainable sodium-ion batteries that can compete with lithium-ion for grid-scale energy storage.

COMMERCIALISATION METRICS

Novelty

4/10

The research appears to be an incremental mechanistic study using standard spectroscopic techniques (NMR/EPR) to characterize a known material class (porous carbon) for sodium-ion batteries. While scientifically sound, it lacks the platform-enabling or paradigm-shifting characteristics required for a high novelty score in deep-tech investing.

Commercial Potential

4/10

The paper focuses on fundamental mechanistic characterization using high-end spectroscopy rather than demonstrating a scalable manufacturing process or a specific battery cell performance metric. While the insights could optimize future materials, the work is foundational science rather than a direct product or platform technology.

Market Size

9/10

Sodium-ion battery technology targets the massive global energy storage and grid-scale battery markets, offering a lower-cost alternative to lithium-ion for stationary storage and mass-market mobility.

Tech Readiness

3/10

The work is focused on fundamental mechanistic elucidation using advanced spectroscopic techniques (NMR and EPR), indicating basic research rather than prototype development or scaling. It resides at the fundamental science stage (TRL 2-3) intended to inform future material design.

IP Strength

3/10

The paper appears to be fundamental characterization research focused on elucidating scientific mechanisms rather than disclosing a novel, scalable material composition or a specific manufacturing process. While the insights could inform future IP, the work itself is descriptive and mechanistic, offering low immediate defensibility.

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2. Structure-controlled sulfur poisoning and hydrogen-induced regeneration in single Pd nanoparticles probed by nanospectroscopy. 22/50

"Engineering self-healing catalysts to boost efficiency and longevity in hydrogen production and fuel cells."

Main researcher: Elad Gross · elad.gross@mail.huji.ac.il.

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Published in Faraday discussions · 2026 Sep 17

ABOUT THIS RESEARCH

The research investigates how sulfur contaminants deactivate palladium nanoparticles and how hydrogen gas can be used to restore their catalytic activity at a molecular level. By using nanospectroscopy, the team identifies how the specific arrangement of atoms dictates whether a catalyst is poisoned or can be regenerated.

COMMERCIAL ANGLE

This fundamental understanding enables the development of more durable and self-healing industrial catalysts for high-efficiency hydrogen production and fuel cell technologies.

COMMERCIALISATION METRICS

Novelty

7/10

The research offers high technical novelty by utilizing advanced nanospectroscopy to observe atomistic-level structure-property relationships in single nanoparticles, providing mechanistic insights into poisoning and regeneration that go beyond traditional ensemble averaging.

Commercial Potential

5/10

The work offers valuable fundamental insights into catalyst durability and regeneration, which has clear industrial relevance for the hydrogen economy and fuel cells. However, it remains in the fundamental characterization stage without a specific product application or a defined scalable process outlined.

Market Size

4/10

The research is highly specialized fundamental science focusing on atomistic catalyst poisoning mechanisms, which addresses a niche technical bottleneck rather than a broad, high-value market segment. While relevant to the massive hydrogen economy and petrochemical industries, the work is basic research that lacks a direct commercial application or a defined scalable market product.

Tech Readiness

2/10

The paper describes fundamental mechanistic studies using nanospectroscopy at the atomic scale, which is characteristic of basic scientific research rather than applied development. It focuses on understanding chemical phenomena (poisoning and regeneration) rather than demonstrating a functional, integrated catalytic device or process.

IP Strength

4/10

The work focuses on fundamental mechanistic insights into catalyst poisoning and regeneration via nanospectroscopy, which is high-value academic knowledge but lacks the clear application or structural novelty required for strong, broad patent claims.

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